Jordan River Project Water Use Plan

Monitoring Surfing Quality below the Jordan River Generating Station (Year 2)

Reference: JORMON#5

Monitoring Surfing Quality Below the Jordan River Generating Station

Study Period: February – April 2007

RRL Recreation Resources Ltd.

December 2007

Water License Requirements

Jordan
Water Licence Requirement Reporting

Ref Study: JOR-WUP-SURF

Monitoring Surfing Quality Below the Jordan River Generating Station

Study Period: March 3 – 31, 2007
Report Date: December 2007

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Surfers Waiting at Jordan River

Executive Summary

Jordan River continues to be one of the most accessible and popular surf locations on Vancouver Island. It has a long history as a surf location for over 30 years. Part of its popularity is attributable to it being accessible by paved public road and also being located close to the communities of Sooke and Victoria. It is also popular due to the quality of surf.

It is an attractive site, with views across the Strait of Juan de Fuca to the Olympic Peninsula. There are basic campsite facilities located at Jordan River which are maintained by Western Forest Products. There is also a small fenced area with two cabins which are maintained by the Jordan River Surf Club.

The surf season extends between October and March, and surf quality can vary substantially depending on ocean conditions over this period.

A surfing survey conducted over the Water Use Plan (WUP) between December 17th, 2000 and March 16th 2001 indicated that surf conditions are also affected by Jordan River Generating Station discharge. In some conditions high discharges may flatten waves making it more difficult for surfers to catch waves against the current.

The final alternative agreed to by the consultative committee includes provision for a minimum of four (4) weekend days during the month of March when maximum discharge will be limited to $30m^3s^{-1}$ during the day. The potential benefit of this flow will be evaluated during this monitoring program, which is scheduled to occur over a six-year period.

The 2007 survey was conducted on March 3, 10, 11, 18, 24, 25 and 31. The 2007 survey had the same format as for 2006; 15 questions and a map on which to record surf use locations, and was conducted using in-person interviews. The focus of the surfing survey is to determine the benefits to the surfing quality at Jordan River as a result of constraints on generation.

A total of 31 surveys were completed during the period March 3 to March 31, 2007 compared with the 40 surveys were completed over the period March 11 – April 1, 2006. The lower number of surveys completed can be attributed primarily to the poor surf conditions which were even poorer in March 2007 than in 2006. There were periods of strong wind during March 2007, however, its direction was usually contrary to that of the surf direction, with the result that the seas were flattened rather than being built up, or, the wind created confused seas which did not build waves of any surfable size or quality.

2007 Results Summary

During this monitoring period, flows from the Jordan River Generating Station were 100% constrained on March 10 and 11 and surf users were present and were surveyed. Flows were partially constrained on March 24 and surf users were present and were surveyed.

On March 3, 18, 25 and 31, surf user monitoring occurred but no surf users were present.

Key observations of the 2007 Monitoring Program are summarized below:

- The primary users of Jordan River continue to be board surfers.
- 42% of surfers rated their experience as poor to fair, and 58% rated the experience as good to excellent.
- Half of the respondents, 50%, had surfed for less than five years, suggesting overall lower levels of experience than in the 2006 surf user survey, where 66% had more than 5 years experience.
- The maximum number of users in the water during the on-site survey times was 17.
- Shakies Beach and Points Left were the most popular surfing areas.
- 87% of users lived within 100 km of Jordan River. This correlates well with the amount spent. 55% of surfers spent $10-$25 for each surf day, including transportation, meals and accommodation. This would cover fuel and one or two meals for a surfer driving out to Jordan River from Sooke or Victoria.
- 82% of surf users observed that the water discharge from Jordan River did not affect their surf experience. During this survey period, flows from the Elliot Head Pond were 100% constrained (no flow) on two days when surf user surveys were completed, and constrained on one day when surf user surveys were completed.
- The benefits of the unconstrained flow, which was expected to reduce surf quality through increased current and reduced water quality, was difficult to assess during this monitoring period. This was due in part to the small surf conditions, and was also due to low numbers of surf users and the fact that 50% of the users had surfed for less than five years. Surfers with limited experience are expected to have less ability to assess the impacts of river flow and other factors on their surfing experience than experienced surf users.

1.0 2007 Survey Objective and Scope

The objective of the surf-monitoring program is to assess the performance of the Water Use Plan decision to constrain maximum discharge opportunistically during the surfing season, using surf quality as the performance measure.

The monitoring program will be assessed near the mouth of the Jordan River in the intertidal zone where surfing is prevalent. It will evaluate the surf quality of waves that are particularly vulnerable to Jordan River Generating Station operations, based on the response of experienced surfers.

Surf quality will be evaluated under two conditions: constrained and unconstrained generation discharge, alternating over the review period of six years. 2007 was an unconstrained discharge observation period. 2006 was a constrained discharge observation period.

The objectives of the Jordan River surf users monitoring program are to:

1. Develop baseline information on daily surf use for high use weekends.
2. Determine user residency and activity preferences.
3. Define how often individuals surf at Jordan River on an annual basis.
4. Assess the correlation between discharge from the generating station and surf quality.
5. Record weather and tide information during the periods of surf survey administration.
6. Establish a monetary value for a surfer's day.

2.0 Survey Methodology

The survey of Jordan River surf users commenced March 10, 2007 and was delivered on site through in-person interviews. The surf user survey was explained to the user and was completed by either Erin Prescott or Jeremy Webb.

Survey monitoring was conducted on March 3, 10, 11, 18, 24, 25 and 31. On only three of these monitoring periods were there surf users at Jordan River, on March 10, 11 & 24th. There was little or no surf on March 3, 18, 25 and 31.

Surf conditions were monitored off-site through web sites and through contact with an experienced surfer and long-term resident living in Sooke, Rick Gillie, who undertook analysis of weather forecasts and provided surf forecasts for Jordan River.

The surf user survey is designed to measure the factors contributing to the surf users experience at Jordan River so as to determine how constraints on power generation affected surfing quality.

There is a short 'window' where the surf user can be interviewed, after they come out of the water, before they wish to get changed and warm up. The length of time it took to complete each survey appeared appropriate, with no users objecting to the survey being too long.

The survey questions were straightforward and generated only normal requests for clarification. Refer to Appendix A to view the Surf Users Survey template.

In addition to the surf user survey, the number of surf users in the water and their locations were observed and recorded at approximately half hour intervals. Weather observations were also made and recorded at approximately one-hour intervals.

Weather information was used to help determine the surf conditions at Jordan River. Particularly useful was the on-line weather stations located at Gordon's Beach in Sooke: http://bigwavedave.ca/GordonsWSRT2.htm. Also used were Environment Canada's Marine Weather forecasts and stations providing wind direction for Juan de Fuca Strait: http://www.weatheroffice.gc.ca/marine/marine_e.html?C-WSP, and a popular site: http://www.bigwavedave.ca/latest.php?dir=Tides2&file=Tidal used by windsurfers and surfers. Also useful is the US-NOAA coastal waters forecasts: http://weather.noaa.gov/cgi-bin/fmtbltn.pl?file=forecasts/marine/coastal/pz/pzz131.txt. which provides weather information for Juan de Fuca Strait.

3.0 Survey Results

A total of 31 surveys were completed during the 2007 monitoring period, significantly less than the 116 surveys collected in 2000/2001 and less that the 40 surveys completed in 2006.

All survey forms were completed correctly. None of the surf users were surveyed more than once.

3.1 User Group Profile

User Activities

The survey records the type of surf use.

#3 – Type of surf use activity										
Board		**Body**		**Boogie**		**Windsurf**		**Kayak Surf**		
Male	Female	Male	Female	Male	Female	Male	Female	Male	Female	
										March 3
11	3									March 10
3	1							1		March 11
										March 18
8	4									March 24
										March 25
										March 31
22	**8**							**1**		**Totals**

The majority of surf users at Jordan River are board users and are male. This is consistent with the 2006 survey which also indicated board users were dominant. There were no surf users on March 3, 18, 25 & 31 due to almost flat calm conditions.

Monitoring was not done on March 4 and 17 as a review of weather information forecast calm conditions.

3.2 Surf User Experiences

A series of four questions was used to differentiate the elements contributing to surf users experiences.

Getting on to a wave in order to ride it is one of the fundamental challenges of surfing, and is a skill that develops with knowledge and experience, including familiarity with an area and its characteristics.

#4 - How would you rate the conditions for catching the waves?								
Poor		Fair		Good		Excellent		
Male	Female	Male	Female	Male	Female	Male	Female	
								March 3
1		8	3	2				March 10
2		2	1					March 11
								March 18
4	2	2		1	3			March 24
								March 25
								March 31
7	**2**	**12**	**4**	**3**	**3**			**Totals**

As occurred in 2006, the surf was unusually poor during this year's survey period, (a time of year when good surfing conditions are normally expected). Most surf users felt conditions for catching waves were poor to fair. Over the survey period, 29% of surfers rated conditions as poor, 52% as fair, 19% as good and 0% as excellent.

The higher the wave, the better the ride. Waves were small again in 2007, and most surf users gave wave height a poor to fair rating.

#5 - How would you rate the wave height?								
Poor		Fair		Good		Excellent		
Male	Female	Male	Female	Male	Female	Male	Female	
								March 3
2	1	7	2	2				March 10
3			1	1				March 11
								March 18
2	1	5	3	1				March 24
								March 25
								March 31
7	**2**	**12**	**6**	**4**				**Totals**

Wave heights measured 0 -2 metres over the survey period. 29% of surf users rated wave height as poor, 58% as fair, and 13% as good.

The way in which waves break affects the surfer's ride in several ways and a good wave break is much sought after. There are four surf breaks at Jordan River: Shakies Beach, Outside River Mouth, Points Left and Points Right.

#6 - How would you rate the wave break?								
Poor		Fair		Good		Excellent		
Male	Female	Male	Female	Male	Female	Male	Female	
								March 3
4	1	6		2		1		March 10
1		2	1	1				March 11
								March 18
		3	1	4	3	1		March 24
								March 25
								March 31
5	**1**	**11**	**2**	**7**	**3**	**2**		**Totals**

19% of surf users rated the wave break as poor, 42% rated the wave break as fair, 32% rated it as good, and 7% rated it as excellent.

Once on a wave, the surfer rides it and generally the longer the ride the better.

#7 - How would you rate the length of ride?								
Poor		Fair		Good		Excellent		
Male	Female	Male	Female	Male	Female	Male	Female	
								March 3
3	2	7	1	1	1			March 10
2		1		1	1			March 11
								March 18
3	1	2	1	2	2	1		March 24
								March 25
								March 31
8	**3**	**10**	**2**	**4**	**4**	**1**		**Totals**

35% of surfers rated the ride length as poor, 38% as fair, 25% as good and 3% as excellent. The majority, 73% rating the ride length as poor to fair is consistent with the surf conditions present during the survey period.

In an attempt to gain a rating from surf users that focused less on specific elements and more on the combination of elements contributing to their experience, surf users were asked to rate their overall experience while out on the water.

#8 - How would you rate the quality of your surfing experience today while you were in the water?								
Poor		Fair		Good		Excellent		
Male	Female	Male	Female	Male	Female	Male	Female	
								March 3
	2	6		4		2		March 10
2		1				1	1	March 11
								March 18
		2		5	2	2	1	March 24
								March 25
								March 31
2	**2**	**9**		**9**	**2**	**5**	**2**	**Totals**

42% of surfers rated their experience as poor to fair, and 58% rated the experience as good to excellent, suggesting that while surf conditions were unremarkable, they provided conditions that pleased most users.

Experienced surfers typically check the conditions at Jordan River before going out by accessing on-line weather information and by relating weather information to what surf conditions at Jordan River will likely be. The use of local, independent weather sites appears to be increasing, and may be having an influence on surf user patterns.

The majority (77%) of surf users surveyed at Jordan River in 2007 had 0-4 years experience, suggesting most experienced surf users may not have been using the site during the survey period as they knew the surf heights would be low.

3.3 Influence of River Flow

For the prior March 1–14, 2007, flows from the Jordan River dam were 100% unconstrained i.e. no water was released. For the period March 15 - 30, 2007, flows from the Jordan River dam were constrained to approximately $33m^3s^{-1}$. During the survey period, surf users were asked if they noticed the flow from the Jordan River.

#9 - Did you notice if the water discharge from the Jordan River affected your water activity?				
Yes		No		
Male	Female	Male	Female	
				March 3
8	1	3	2	March 10 (no flow)
2	1	2		March 11 (no flow)
				March 18
1	3	7	1	March 24 (constrained flow)
				March 25
				March 31
11	**5**	**12**	**3**	**Totals**

On March 10 & 11, when flows were fully constrained, 63% of surf users indicated they did notice river discharge affecting their experience while 36% of surf users indicated they did not notice river discharge affecting their experience while. On March 24, when constrained flow was present, while 33% indicated they did notice the river discharge, while 66% did not notice the discharge affecting their experience.

The 2007 results appear inconsistent with 2001 results when surfers were correct 57% of the time in identifying whether or not there is discharge and with 2006, when flows were fully constrained (no flow) during the survey period and 82% of surf users observed that the water discharge from the Jordan River did not affect their surf experience. The 2007 sample size is small, and user experience is lower than in 2001 and 2006 which may partially explain the inconsistency.

3.5 Surf User Numbers

During the 2007 survey period, the maximum number of surfers in the water at any one time was 13. In 2006 there was a maximum of 17 surfers in the water at any one time

In 2001, the maximum number of surfers in the water at any point in time, during the on-site surveys, was 22 surfers. Although there has been a significant increase in use at Jordan River since the first surf user surveys were completed in 2001, this increase was not observed during the 2006 or 2007 monitoring periods.

Number of surf users in the water: March 3, 10, 11, 18, 24, 31 - 2007													
		0900	0930	1000	1030	1100	1130	1200	1230	1300	1330	1400	1430
Mar. 3	Pts R		0		0		0						
	Pts L		0		0		0						
	Out		0		0		0						
	Shake		0		0		0						
Total			**0**		**0**		**0**						
		0900	0930	1000	1030	1100	1130	1200	1230	1300	1330	1400	1430
Mar. 10	Pts R	6	6	6	2	2	2	1	1				
	Pts L	0	0	0	0	0	0	2	0				
	Out	0	1	1	0	7	7	0	1				
	Shake	7	4	5	5	2	2	0	0				
Total		**13**	**11**	**12**	**7**	**11**	**11**	**3**	**2**				
		0900	0930	1000	1030	1100	1130	1200	1230	1300	1330	1400	1430
Mar. 11	Pts R		0			1	4	1	2				
	Pts L		0	1									
	Out		0										
	Shake		1	(1)	(1)								
Total			**1**	**2**	**1**	**1**	**4**	**1**	**2**				
		0900	0930	1000	1030	1100	1130	1200	1230	1300	1330	1400	1430
Mar. 18	Pts R		0		0		0						
	Pts L		0		0		0						
	Out		0		0		0						
	Shake		0		0		0						
Total			**0**		**0**		**0**						
		0900	0930	1000	1030	1100	1130	1200	1230	1300	1330	1400	1430
Mar. 24	Pts R	6	2	7	7	8	5	3	3				
	Pts L												
	Out												
	Shake	4	4	5	2	1	1						
Total		**10**	**6**	**12**	**9**	**9**	**6**	**3**	**3**				
		0900	0930	1000	1030	1100	1130	1200	1230	1300	1330	1400	1430
Mar. 25	Pts R			0		0							
	Pts L			0		0							
	Out			0		0							
	Shake			0		0							
Total				**0**		**0**							
		0900	0930	1000	1030	1100	1130	1200	1230	1300	1330	1400	1430
Mar. 31	Pts R	0		0		0		0					
	Pts L	0		0		0		0					
	Out	0		0		0		0					
	Shake	0		0		0		0					
Total		**0**		**0**		**0**		**0**					

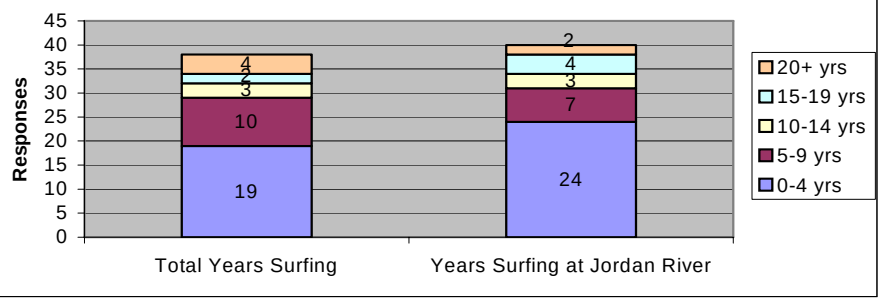
Notes:

1. Pts R = Points Right/Pts L = Points Left/Out = Outside/Shake = Shakies Beach
2. Numbers in ( ) are kayak surfers, numbers in [ ] are boogie board surfers, and numbers outside of the brackets are board surfers.

3.6 User Experience

For some of the surfers surveyed, it was their first time surfing anywhere. At the other end of the spectrum were a limited number of surfers who have been surfing for over 20 years. 77% of the surf users surveyed at Jordan River had been surfing between 0-4 years. For 10 surf users, it was their first time surfing at Jordan River. Overall, surfer experience is low, with 61% having 0-4 years surfing experience.

Figure 1 2007 Surf User Experience



3.7 Number of Surfing Days

The number of days individuals surf at Jordan River each year varied from once per year, to over 60 times per year.

Number of days surfing at Jordan River per year					
0-9	10-19	20-29	30-39	40-59	60+
12	3	4	5	2	4
39%	10%	13%	16%	7%	13%

3.8 User Residency

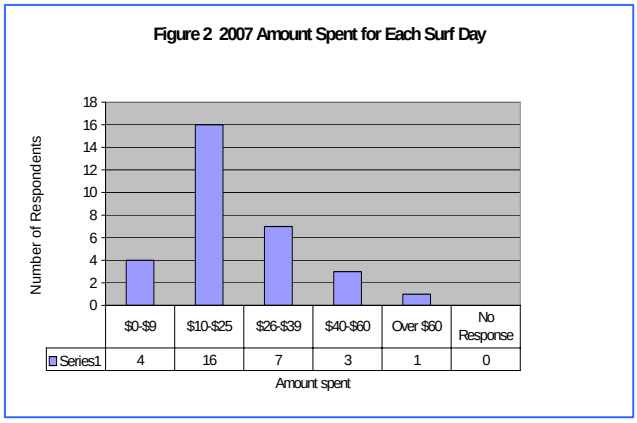
Surf users were asked where they lived.

Residency of surfers users at Jordan River				
Within 50 km	50-100 km	100+ km	Other Canadian Area	Non Canadian
17	10	5	0	0
36%	50%	14%	0%	0%

36% of surf users lived within 50 km of Jordan River, 50% lived 50-100 km, 14% live 100 km or greater from Jordan River. All individuals were from BC.

3.9 Amount Spent

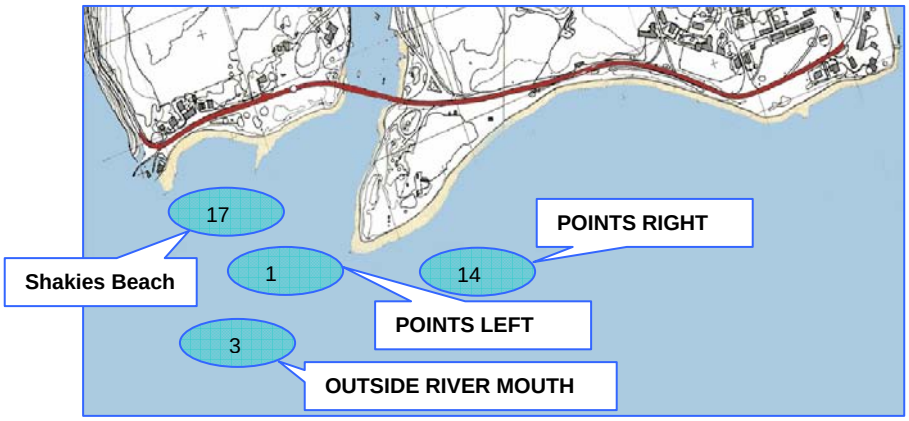
Surf users were asked the total spent for each surf day including transportation, meals and accommodation (including taxes).



13% of surf users indicated that they spent approximately $0-$9 on each surf day, including transportation, meals and accommodation (see Figure 2). 52% of surf users spent $10 - $25 for each surf day, 23% indicated they spent $26-$39, 10% indicated they spent $40-60, and one individual indicated he spent over $60 per surf day.

3.9.1 Surfing Location

Surfers were asked where they surfed during their time at Jordan River. Some surfers used two locations during their time in the water. The most used areas for surfing were Shakies Beach and Points Right. The map below shows the number of surfers using each area over the survey period.



4.0 Summary of Observations

The prime surfing season continues to be November to March, with opportunities for good surfing also occurring outside of the high season, i.e. September and October.

The surf user surveys are designed to encompass environmental influences upon surf user experiences at Jordan River. These influences included tides, swell from offshore winds and near shore wind and wave conditions.

The 2007 study collected 31 surveys during the monitoring period (8 less than in 2006). This small sample size, combined with poor surf conditions and lower overall surfer experience does not justify extensive analysis. However, a number of observations can be made:

- Primary users of the Jordan River surf area continue to be board surfers and are male. There are the occasional surf kayakers.
- 42% of surfers rated their experience as poor to fair, and 58% rated the experience as good to excellent, suggesting that while surf conditions were unremarkable, they provided conditions that pleased the majority of users.
- 77% of the surf users surveyed at Jordan River had been surfing between 0-4 years. For 10 surf users, it was their first time surfing at Jordan River. Overall, surfer experience is low, with 61% having 0-4 years surfing experience.
- The maximum number of users in the water during the 2007 survey was 13.
- The survey results show a group of experienced local surfers continue to surf Jordan River on a regular basis. 36% of surf users lived within 50 km of Jordan River, 50% lived 50-100 km, 14% live 100 km or greater from Jordan River. All individuals were from BC
- Shakies Beach and Points Right were the most popular surfing areas in 2007. In 2001 Points Right and Points Left were most popular. In 2006 Shakies Beach and Points Left were most popular.
- 86% of users lived within 100 km of Jordan River. This correlates well with the amount spent. 88% of surfers spent $0-39 for each surf day, including transportation, meals and accommodation. This amount would cover fuel and one or two meals for a surfer driving out to Jordan River from Sooke or Victoria.
- 48% of surf users indicated they did not notice river discharge affecting their experience while 52% of surf users indicated they did notice river discharge affecting their experience, despite this being an unconstrained flow year.

5.0 Recommendations

The low number of surf users surveyed at Jordan River over the past two seasons has been disappointing. Poor surf conditions (i.e. small waves) occurred in 2006 and in2007 and attracted low numbers of surf users.

The survey format continues to work well and should continue to be used in 2008 and should be conducted during the same period and for every weekend in March so as to capture as large a sample size as possible.

The survey work should continue to make use of weather sites and on-line forecasting tools.

The 2008 surf user report should include a detailed summary section containing tables of the most important metrics (conditions, experience, influence of river flow, etc.) collected over the 4 periods of data collection (2001/2006/2007/2008) so as to enable comparisons between constrained flow and unconstrained flow conditions.

Appendix A – 2007 Surf Survey

Attention Jordan River Surf Users!

As part of BC Hydro's Jordan River Water Use Plan Monitoring Program, a survey of Jordan River surf users is being undertaken. A hydroelectric dam controls the amount of water that flows in the Jordan River. Of particular interest is how the amount of water discharge from the hydroelectric dam may affect surfing conditions.

BC Hydro has contracted an independent consulting company, RRL Recreation Resources Ltd. (RRL) to undertake the surf user survey. If you have any questions regarding the survey, you can call them, toll free at 1-800-761-5737. Thank you for participating!

Jordan River Water Use Plan - Ocean Surf Recreation User Survey

1. **Date/time:** (Month/Day/Year/time) _____ /_____/_____/_____am/pm
2. **Time in Water:** Time in _____am/pm - Time out _____am/pm
3. **Activity Undertaken:**

   ☐ Board Surf ☐Body surf ☐ Boogie Board ☐ Wind Surf ☐ Kayak Surf

   ☐ *Other (describe)*_____
4. **How would you rate the conditions for catching the waves?**

   ☐ Poor ☐ Fair ☐ Good ☐ Excellent
5. **How would you rate the wave height?**

   ☐ Poor ☐ Fair ☐ Good ☐ Excellent
6. **How would you rate the wave break?**

   ☐ Poor ☐ Fair ☐ Good ☐ Excellent
7. **How would you rate the length of ride?**

   ☐ Poor ☐ Fair ☐ Good ☐ Excellent
8. **How would you rate the quality of your surfing experience today while you were in the water?**

   ☐ Poor ☐ Fair ☐ Good ☐ Excellent
9. **Did you notice if the water discharge from the Jordan River affected your water activity?**

   ☐ Yes ☐ No

   If answering yes to Question 9, please describe how your activity was affected.

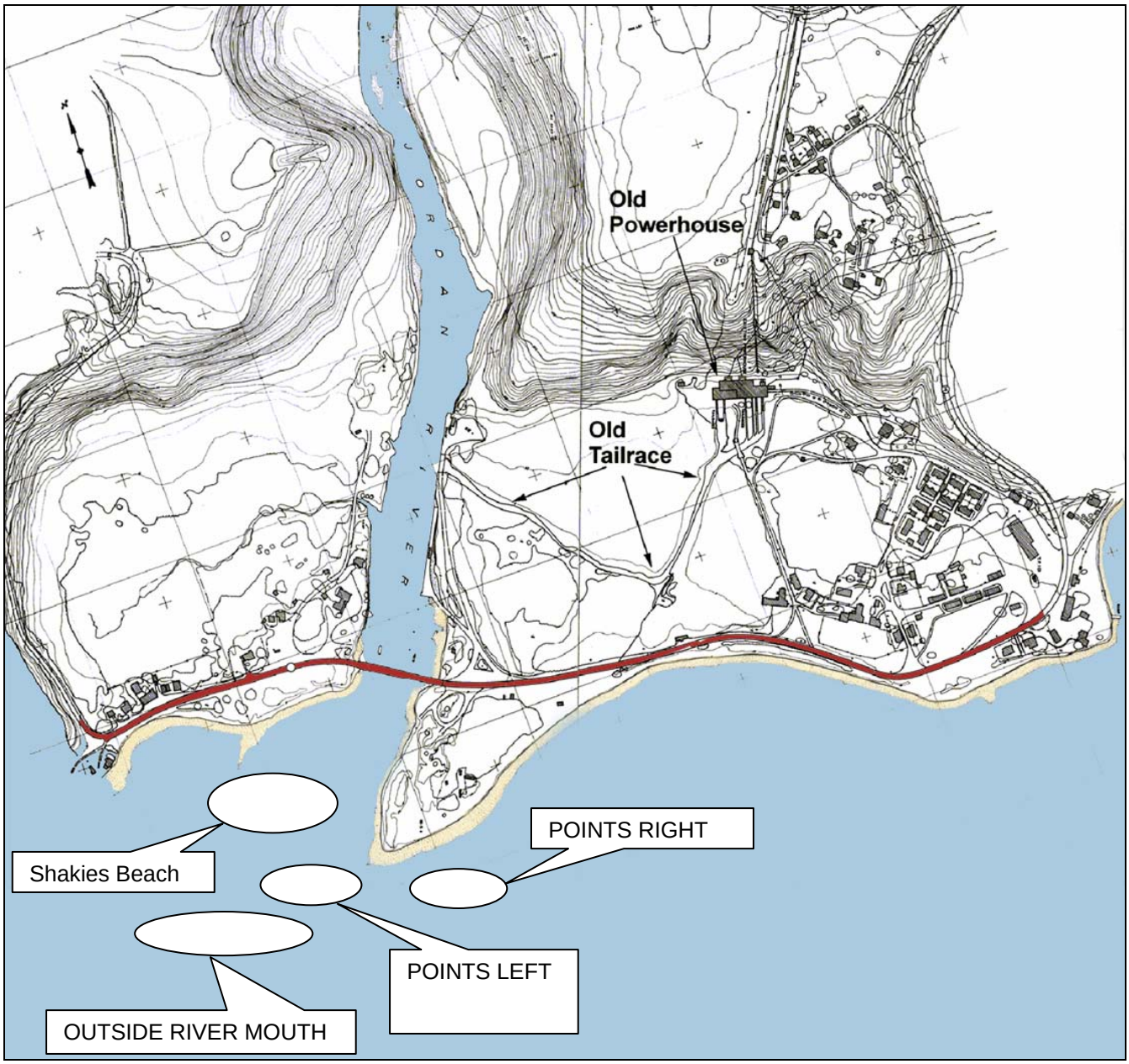
   _____
10. **Have you been interviewed before?**

    ☐ Yes ☐ No
11. **How many years have you been surfing at Jordan River?** _____
12. **How many years have you been surfing?**_____
13. **On average, how many days do you surf at Jordan River each year**?_____
14. **Where do you live**?

    ☐ Within 50 km ☐ 50-100 km ☐ 100+km ☐ Other Canadian area _____ ☐ Non-Canadian_____
15. **How much do you spend in total for each surf day, including transportation, meals and**

accommodation (including taxes)? ☐ $0-$9 ☐ $10-$25 ☐ $26-$39 ☐ $40 - $60 ☐ over $100

On the map below, please circle the location where you surfed*:* *(Point Rights, Point Lefts, Shakies Beach, Outside Jordan River Mouth)*



Appendix B - Discharge Data For Jordan River Dam

The tables below list flow data for the dates on which surf users were present at Jordan River during the 2007 monitoring period.

Dam	Date	Hour	m^3s^{-1}
JOR	3/10/07	7	0
JOR	3/10/07	8	0
JOR	3/10/07	9	0
JOR	3/10/07	10	0
JOR	3/10/07	11	0
JOR	3/10/07	12	0
JOR	3/10/07	13	0
JOR	3/10/07	14	0
JOR	3/10/07	15	0
JOR	3/10/07	16	0
JOR	3/10/07	17	0
JOR	3/10/07	18	0
JOR	3/11/07	7	0
JOR	3/11/07	8	0
JOR	3/11/07	9	0
JOR	3/11/07	10	0
JOR	3/11/07	11	0
JOR	3/11/07	12	0
JOR	3/11/07	13	0
JOR	3/11/07	14	0
JOR	3/11/07	15	0
JOR	3/11/07	16	0
JOR	3/11/07	17	0
JOR	3/11/07	18	0

Dam	Date	Hour	m^3s^{-1}
JOR	3/24/07	7	33.649
JOR	3/24/07	8	33.306
JOR	3/24/07	9	33.306
JOR	3/24/07	10	33.653
JOR	3/24/07	11	33.31
JOR	3/24/07	12	33.649
JOR	3/24/07	13	33.302
JOR	3/24/07	14	33.298
JOR	3/24/07	15	33.641
JOR	3/24/07	16	33.298
JOR	3/24/07	17	33.302
JOR	3/24/07	18	33.649

Appendix C - Tidal Data

Tides for the survey periods when surfers were recorded at Jordan River: March 10, 11 and 24th. Data from Fisheries and Oceans Canada Port Renfrew Tide Station:

Port Renfrew (Station #8525)

2007-03-10 (Saturday)		
Time	**Height**	
PST	(m)	(ft)
03:31	**2.9**	9.5
11:08	**1.2**	3.9
17:15	**2.1**	6.9
21:21	**1.8**	5.9

2007-03-11 (Sunday)		
Time	**Height**	
PST	(m)	(ft)
04:10	**2.9**	9.5
12:18	**1.2**	3.9
19:06	**2.0**	6.6
21:51	**1.9**	6.2

2007-03-24 (Saturday)		
Time	**Height**	
PDT	(m)	(ft)
04:38	**3.2**	10.5
12:19	**0.7**	2.3
18:56	**2.2**	7.2
22:48	**1.9**	6.2

Appendix D – Weather Conditions

Date	Time	Wind Direction	Wind Speed (Km/Hr)	Temp Air (Degrees Centigrade)	Estimated Wave Height (Feet)			
					Shakies	Outside River Mouth	Points Left	Points Right
March 3	0948	N	2-6	8	1.0	1.0	0.0	1.0
	1056	N	5-10	10	1.0	0.0	0.0	1.0
	1200	N	6-8	9	1.0	1.0	1.0	1.0
	1300	N	8-10	9	1.0	1.0	0.0	1.0
March 10	0900	E	0-4	8	3.0	3.0	3.0	3.0
	1000	n/a	0	9	3.0	3.0	2.0	2.0
	1100	n/a	0	8	2.0	2.0	2.0	2.0
	1215	NW	6	8	3.0	3.0	3.0	3.0
March 11	1000	N	0-4	10	2.0	0	1.5	1.5
	1110	N	0-4	10	2.0	0	1.0	1.5
	1200	N	0-4	10	1.0	0	1.0	1.0
March 18	0930	WSW	10-14	11	1.0	0	2.0	2.0
	1030	WSW	16-20	14	2.0	2.0	2.0	2.0
	1130	W	11-14	12	2.0	0.0	2.0	2.0
March 24	0915	NE	0-4	10	3.0	0.5	1.5	3.0
	1000	n/a	0	9	1.0	0.0	0.0	0.0
	1100	n/a	0	9	2.0	0.5	0.5	2.0
	1200	n/a	0	9	1.0	0.0	0.0	1.0
March 25	0930	NW	9	8	1.0	1.0	1.5	1.5
	1030	W	9	8	0.5	0.5	0.5	0.5
March 31	0930	W	8-10	8	1.0	1.0	1.0	1.0
	1030	W	6-8	11	2.0	2.0	2.0	2.0
	1130	W	18-20	14	2.0	2.0	3.0	3.0
	1230	W	15-18	12	3.0	2.0	3.0	2.0